Statistical Scraping
Interest Group

WEB-FOSS-NL

Statistical Scraping explored in the Netherlands

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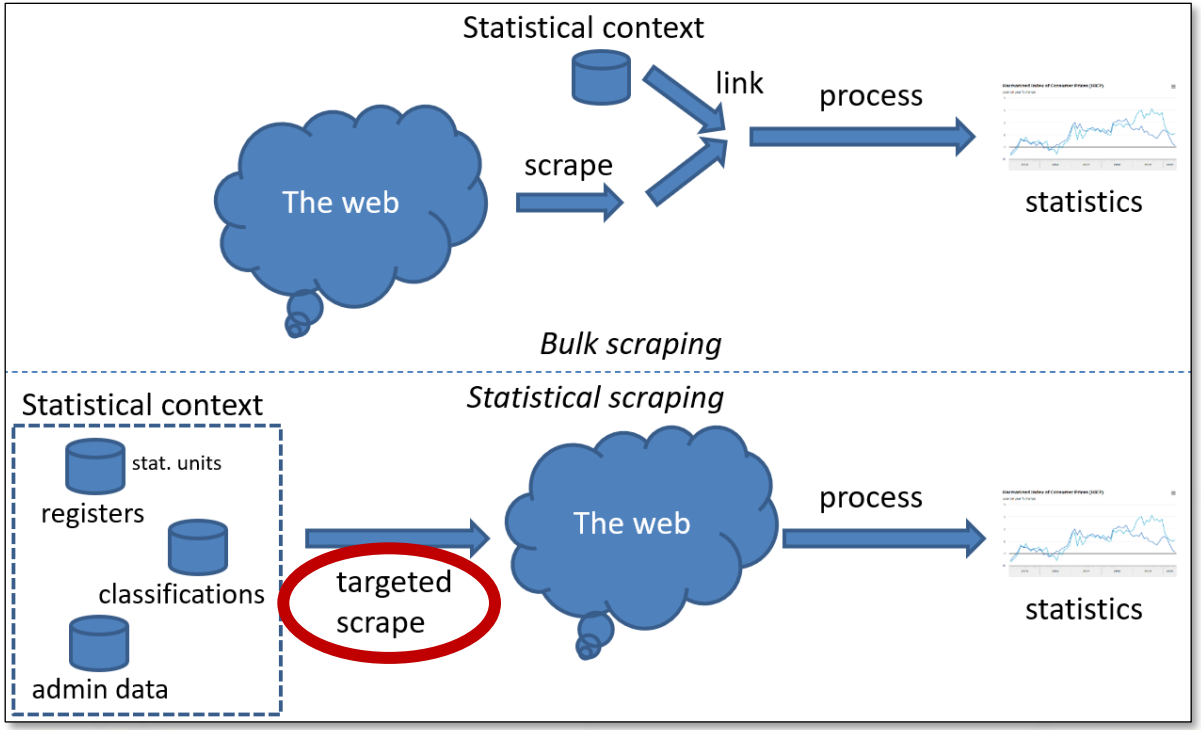
SSIG2 – Statistical Scraping Interest Group meeting 2, 15-16 Apr. The Hague

Web data in official statistics: why?

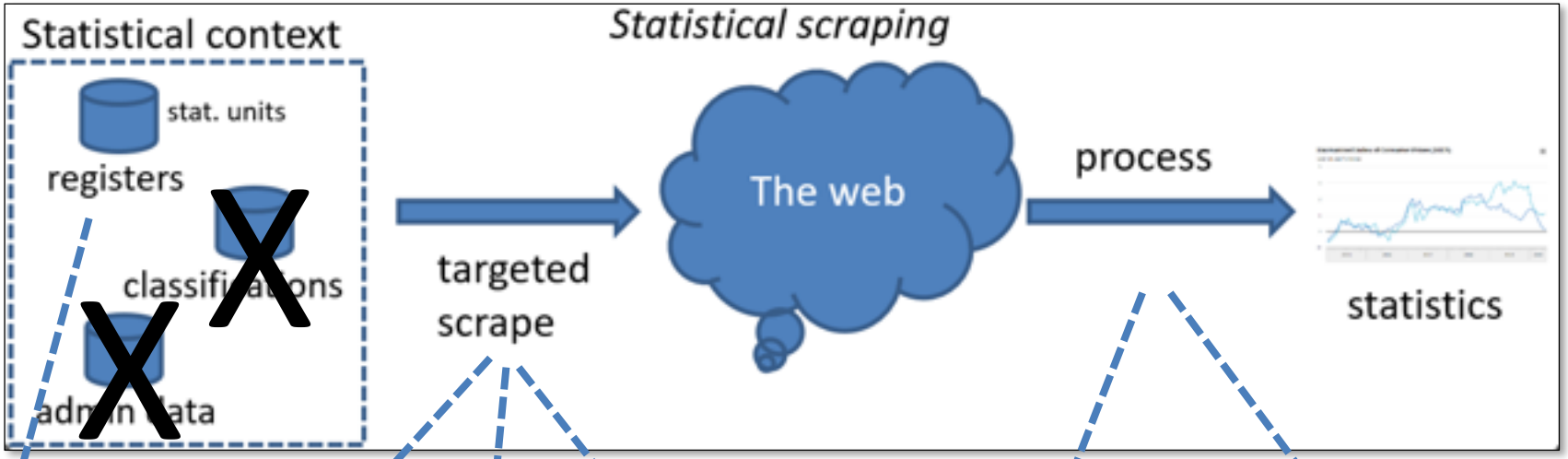
- ***Less surveys***: get it from the web instead of asking
- ***Faster***: use uptodate web data instead of slow media
- ***New indicators***: measure phenomena that can not be measured traditionally (e.g. internet economy)
- ***Improving*** administrative sources / register data
- ***Discover*** missing entities

Statistical scraping: high level view (2024)



[Paper Q conference (2024)]

Statistical Scraping applied to job vacancies



Statistical business register (SBR)

Sample (WP2)

URL finding (WP2)

Focused scrape (WP3)

Interpret Text mining / LLMs (WP3)

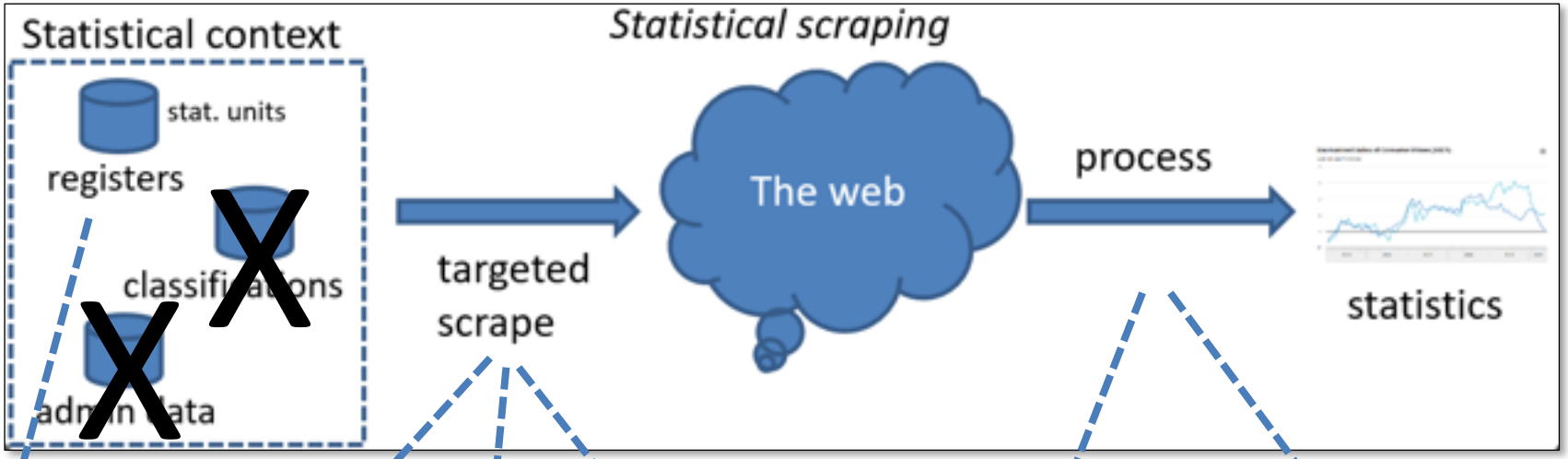
Experimental stats (WP3)

Validation

JVS

Official stats

Statistical Scraping concept applied to JV (4)



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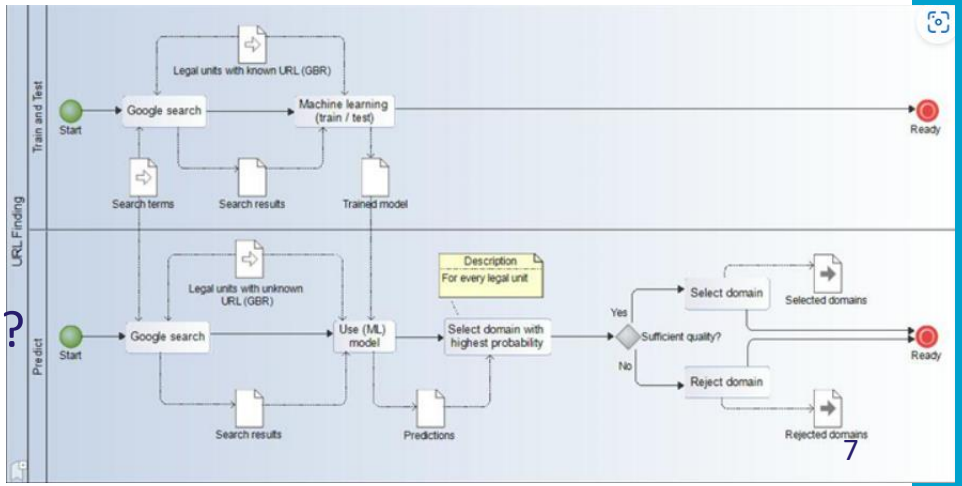
Building blocks in progress: URLfinding

Finding website(s) of enterprises in our business register

Dutch business register (roughly):
- 700.000 legal units with known URL (but may be incorrect)
- 2.400.000 legal units with unknown URL

URL finding: plans (last meeting)

- https://github.com/SNStatComp/urlfinding
- Generic software for finding enterprise websites using Google search
- 5 years old; used in production
- Needs refactoring, retraining of ML model Google API
- Other search engines? (DuckduckGo?)
- Open web search project EU?: https://openwebsearch.eu/
- Other web sources?
  spatial data (maps), like in WIN
- Your URL finding software (ideas)?



URL finding: query reduction

- Originally 6 Google search queries per legal unit ("enterprise")
- Challenge: reduce #queries, without losing quality:
  - How to optimally use contact information, CoC id, address in query?
  - greedy strategy: on a test set of 1000 enterprises with **known URL** (greedy: leave one out and analyse results)
  - => address and 'contact' appeared to be redundant

[1] {term: [TradeName, ' ', Location, ' ', "(contact | contactgegevens | kvk | adres)"], orTerm: [LegalName]}

[2] {term: [TradeName, ' ', Postalcode, ' ', "(contact | contactgegevens | kvk | adres)"], orTerm: [LegalName]}

[4] {term: [TradeName], orTerm: [LegalName]}

Location = address + postalcode
Redundant with Postalcode?
Nevertheless adds useful results
=> further research

URL finding: software improvements

- Two new releases: v.0.0.4 and v0.0.5 (see release notes)
- ***Model maintenance***:
  - Google search moves on continuously => model *retraining* needed
- ***DuckDuckGo*** implemented:
  - However no API and blocking -> only for small data sets
- Used for 1st WEB-FOSS-NL **test run**:
  - Selected 23.000 legal units with known (valid) URL from SBR
  - URL finding for a sample of 6.600 legal units => 2.300 hits
  - Google limit 10.000 / day => two days run; total costs: < Euro 200
  - Data delivered for next steps

Building blocks in progress: Focused scrape

Goal (last meeting)

- **Scrape only** (all) the pages of a website that are likely to contain a **certain type of content**
- Our focus in this project: **Online job advertisements** (OJAs) of enterprises

- **Input**: URL of an enterprise, e.g., https://www.cbs.nl
- **Output**: Set of pages with potential OJA texts
- Experimental use of LLMs/ AI to distinguish OJA's from NOJA's

1st Proof of Concept run March 2026

Input from URL finding: 2335 unique base-URL's of known NL units

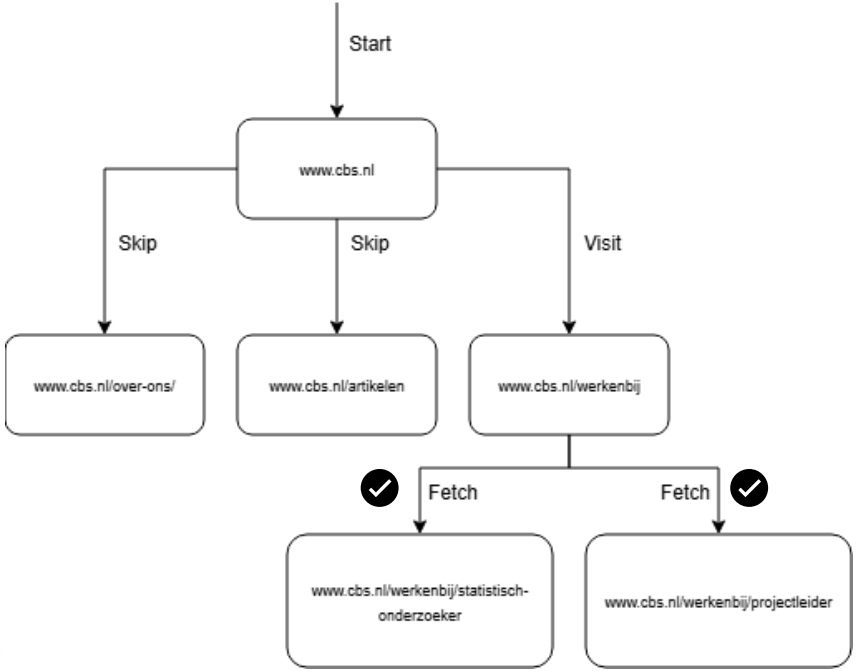
(Regex) Keywords

werk(en)?-?bij
vacature(s)?
job(s)?
career(s)?
cari(e|è)re
collega
versterk

Results

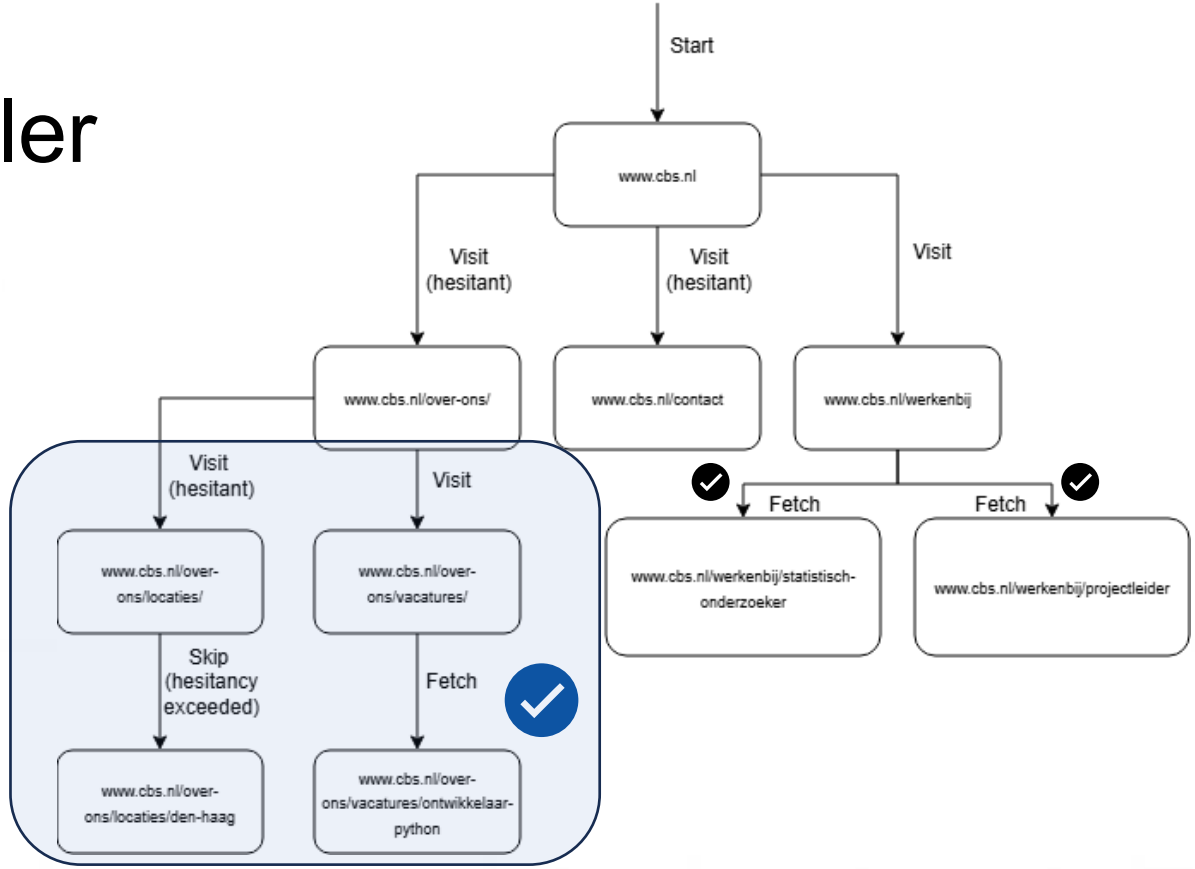
Base-URLs	Processed	Visited	Got results	Pages
2335	2335	No: 221 - Got stuck: 5 - Request failure: 217		
		Yes: 2114	No: 775	
			Yes: 1339	6773

Usual Crawler



Hesitant Crawler

Digs **deeper** than regular keyword-based crawler



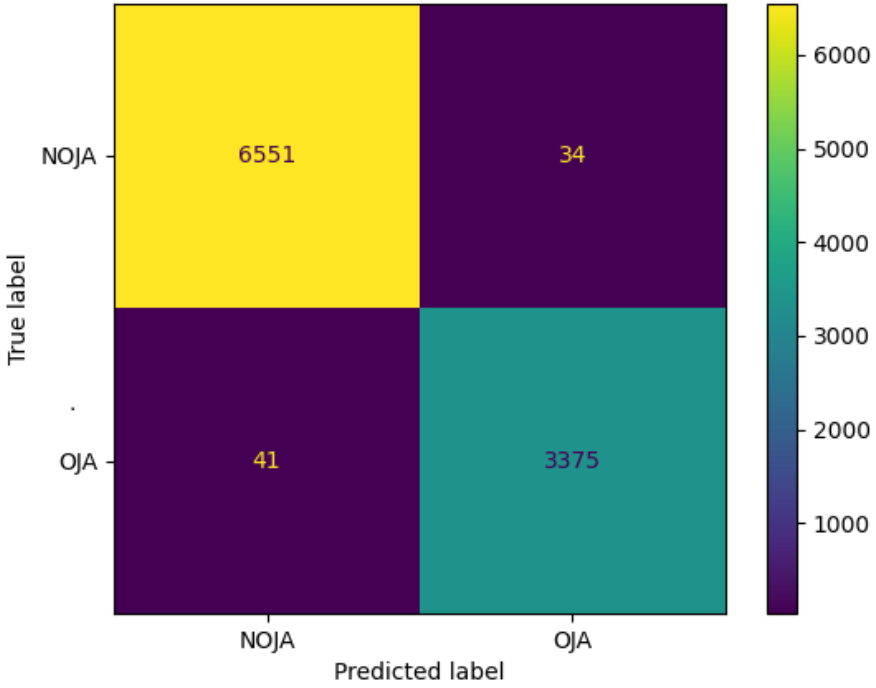
Building blocks in progress: Text mining / LLMs

Model and data for PoC

- Embedding and classifier training (instead of LLM prompting)
  - Pretrained embedding model
  - Own training of classifier: Is this an OJA yes or no?
- Initial training/ testing data:
  - From Eurostat Web Intelligence Hub:
    ~10,000 Dutch-language OJA (positives: OJA)
  - ~20000 Dutch-language wikipedia articles (negatives: NOJA)

Initial Classifier

Tested on initial testing data

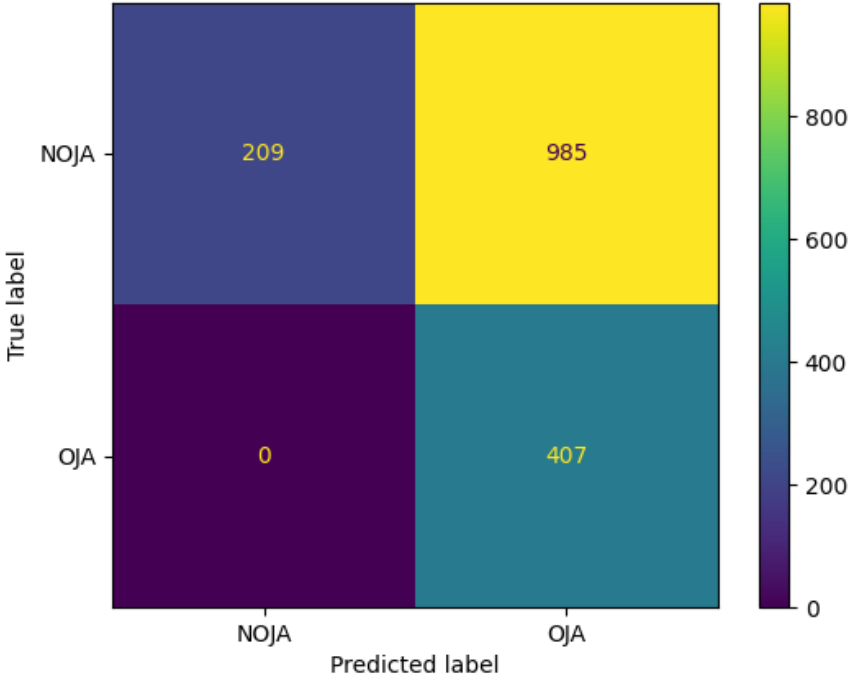


Initial Classifier

Tested on 1600 newly labelled data from PoC-scrape

→Model needed finetuning

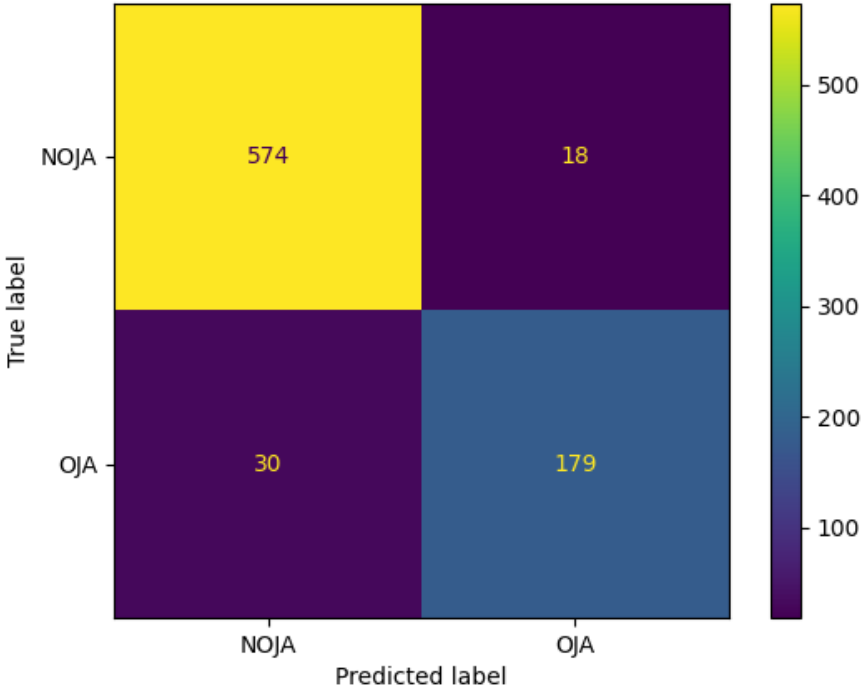
→using newly-labelled data



Current Classifier

Much better!

Needs additional validation



PoC Results

- Scrape still leads to a few weird cases, more debugging necessary
- Next steps after first release:
  - More mature Scraper (faster, async, JS)

- Classifier needed additional finetuning with more typical NOJA's
  - Extend labelling with more detail
  - Distinguish OJA's from e.g. volunteering internships etc
- Finetuning with newly labelled data from PoC helps!

- Next PoC: Run bigger-scale!